

MongoDB Lab – Day 5 (Indexing & Sorting)

Schema { emp_id, name, department, designation, city, salary, experience }

Initial Data:

Emp ID	Name	Department	Designation	City	Salary	Experience
101	Amit	IT	Developer	Kolkata	60000	3
102	Neha	HR	HR Executive	Delhi	45000	2
103	Ravi	IT	Tester	Mumbai	50000	4
104	Pooja	Finance	Accountant	Chennai	55000	5
105	Karan	IT	Developer	Kolkata	65000	6
106	Sneha	HR	Manager	Delhi	70000	7
107	Arjun	Finance	Analyst	Mumbai	48000	2
108	Meera	IT	Developer	Bangalore	75000	8
109	Rohit	IT	Team Lead	Pune	90000	10
110	Anjali	HR	Recruiter	Kolkata	42000	1

Inserting Documents:

```
test> use college_CSE_326
switched to db college_CSE_326
college_CSE_326> db.student_CSE_326.insertMany([
... { emp_id: 101, name: "Amit", department: "IT", designation: "Developer", city: "Kolkata", salary: 60000, experience: 3 },
... { emp_id: 102, name: "Neha", department: "HR", designation: "HR Executive", city: "Delhi", salary: 45000, experience: 2 },
... { emp_id: 103, name: "Ravi", department: "IT", designation: "Tester", city: "Mumbai", salary: 50000, experience: 4 },
... { emp_id: 104, name: "Pooja", department: "Finance", designation: "Accountant", city: "Chennai", salary: 55000, experience: 5 },
... { emp_id: 105, name: "Karan", department: "IT", designation: "Developer", city: "Kolkata", salary: 65000, experience: 6 },
... { emp_id: 106, name: "Sneha", department: "HR", designation: "Manager", city: "Delhi", salary: 70000, experience: 7 },
... { emp_id: 107, name: "Arjun", department: "Finance", designation: "Analyst", city: "Mumbai", salary: 48000, experience: 2 },
... { emp_id: 108, name: "Meera", department: "IT", designation: "Developer", city: "Bangalore", salary: 75000, experience: 8 },
... { emp_id: 109, name: "Rohit", department: "IT", designation: "Team Lead", city: "Pune", salary: 90000, experience: 10 },
... { emp_id: 110, name: "Anjali", department: "HR", designation: "Recruiter", city: "Kolkata", salary: 42000, experience: 1 }
... ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699ad177dfa6af86b628ca0'),
    '1': ObjectId('699ad177dfa6af86b628ca1'),
    '2': ObjectId('699ad177dfa6af86b628ca2'),
    '3': ObjectId('699ad177dfa6af86b628ca3'),
    '4': ObjectId('699ad177dfa6af86b628ca4'),
    '5': ObjectId('699ad177dfa6af86b628ca5'),
    '6': ObjectId('699ad177dfa6af86b628ca6'),
    '7': ObjectId('699ad177dfa6af86b628ca7'),
    '8': ObjectId('699ad177dfa6af86b628ca8'),
    '9': ObjectId('699ad177dfa6af86b628ca9')
  }
}
```

Lab Questions:

Q1. Display all employee records.

Query: db.student_CSE_326.find()

Output:-

```
[
  {
    _id: ObjectId('699ad84d635e9eb4f2628ca0'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('699ad84d635e9eb4f2628ca1'),
    emp_id: 102,
    name: 'Neha',
    department: 'HR',
    designation: 'HR Executive',
    city: 'Delhi',
    salary: 45000,
    experience: 2
  },
  {
    _id: ObjectId('699ad84d635e9eb4f2628ca2'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 50000,
    experience: 4
  },
  {
    _id: ObjectId('699ad84d635e9eb4f2628ca3'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
],
```

```
{
  _id: ObjectId('699ad84d635e9eb4f2628ca4'),
  emp_id: 105,
  name: 'Karan',
  department: 'IT',
  designation: 'Developer',
  city: 'Kolkata',
  salary: 65000,
  experience: 6
},
{
  _id: ObjectId('699ad84d635e9eb4f2628ca5'),
  emp_id: 106,
  name: 'Sneha',
  department: 'HR',
  designation: 'Manager',
  city: 'Delhi',
  salary: 70000,
  experience: 7
},
{
  _id: ObjectId('699ad84d635e9eb4f2628ca6'),
  emp_id: 107,
  name: 'Arjun',
  department: 'Finance',
  designation: 'Analyst',
  city: 'Mumbai',
  salary: 48000,
  experience: 2
},
{
  _id: ObjectId('699ad84d635e9eb4f2628ca7'),
  emp_id: 108,
  name: 'Meera',
  department: 'IT',
  designation: 'Developer',
  city: 'Bangalore',
  salary: 75000,
  experience: 8
},
},
```

```
{
  _id: ObjectId('699ad84d635e9eb4f2628ca8'),
  emp_id: 109,
  name: 'Rohit',
  department: 'IT',
  designation: 'Team Lead',
  city: 'Pune',
  salary: 90000,
  experience: 10
},
{
  _id: ObjectId('699ad84d635e9eb4f2628ca9'),
  emp_id: 110,
  name: 'Anjali',
  department: 'HR',
  designation: 'Recruiter',
  city: 'Kolkata',
  salary: 42000,
  experience: 1
},
},
```

Q2. Create an index on emp_id field.

Query: db.student_CSE_326.createIndex({emp_id:1})

Output: emp_id_1

Q3. Create an index on department field.

Query: db.student_CSE_326.createIndex({department:1})

Output: department_1

Q4. Create a compound index on department and salary.

Query: db.student_CSE_326.createIndex({department:1,salary:1})

Output: department_1_salary_1

Q5. Find employees working in IT department.

Query: db.student_CSE_326.find({department:"IT"})

Output:

```
{
  _id: ObjectId('699ada177dfa6af86b628ca0'),
  emp_id: 181,
  name: 'Amit',
  department: 'IT',
  designation: 'Developer',
  city: 'Kolkata',
  salary: 60000,
  experience: 3
},
{
  _id: ObjectId('699ada177dfa6af86b628ca2'),
  emp_id: 183,
  name: 'Ravi',
  department: 'IT',
  designation: 'Tester',
  city: 'Mumbai',
  salary: 50000,
  experience: 4
},
{
  _id: ObjectId('699ada177dfa6af86b628ca4'),
  emp_id: 185,
  name: 'Karan',
  department: 'IT',
  designation: 'Developer',
  city: 'Kolkata',
  salary: 65000,
  experience: 6
},
{
  _id: ObjectId('699ada177dfa6af86b628ca7'),
  emp_id: 188,
  name: 'Neeraj',
  department: 'IT',
  designation: 'Developer',
  city: 'Bangalore',
  salary: 75000,
  experience: 8
},
{
  _id: ObjectId('699ada177dfa6af86b628ca8'),
  emp_id: 189,
  name: 'Rohit',
  department: 'IT',
  designation: 'Team Lead',
  city: 'Pune',
  salary: 90000,
  experience: 10
}
```

Q6. Display employees whose salary is greater than 60,000.

Query: `db.student_CSE_326.find({salary:{>60000}})`

Output:

```
{
  "_id": ObjectId('699adal77dfa6af86b628ca4'),
  "emp_id": 185,
  "name": 'Karan',
  "department": 'IT',
  "designation": 'Developer',
  "city": 'Kolkata',
  "salary": 65000,
  "experience": 6
},
{
  "_id": ObjectId('699adal77dfa6af86b628ca5'),
  "emp_id": 186,
  "name": 'Sneha',
  "department": 'HR',
  "designation": 'Manager',
  "city": 'Delhi',
  "salary": 78000,
  "experience": 7
},
{
  "_id": ObjectId('699adal77dfa6af86b628ca7'),
  "emp_id": 188,
  "name": 'Meera',
  "department": 'IT',
  "designation": 'Developer',
  "city": 'Bangalore',
  "salary": 75000,
  "experience": 8
},
{
  "_id": ObjectId('699adal77dfa6af86b628ca8'),
  "emp_id": 189,
  "name": 'Rohit',
  "department": 'IT',
  "designation": 'Team Lead',
  "city": 'Pune',
  "salary": 98000,
  "experience": 10
}
}
```

Q7. Sort employees by salary in descending order.

Query: `db.student_CSE_326.find().sort({salary:-1})`

Output:

```
{
  "_id": ObjectId('699adal77dfa6af86b628ca8'),
  "emp_id": 189,
  "name": 'Rohit',
  "department": 'IT',
  "designation": 'Team Lead',
  "city": 'Pune',
  "salary": 98000,
  "experience": 10
},
{
  "_id": ObjectId('699ad84d63be9ebbf2628ca7'),
  "emp_id": 188,
  "name": 'Meera',
  "department": 'IT',
  "designation": 'Developer',
  "city": 'Bangalore',
  "salary": 75000,
  "experience": 8
},
{
  "_id": ObjectId('699adal77dfa6af86b628ca7'),
  "emp_id": 188,
  "name": 'Meera',
  "department": 'IT',
  "designation": 'Developer',
  "city": 'Bangalore',
  "salary": 75000,
  "experience": 8
},
{
  "_id": ObjectId('699ad84d63be9ebbf2628ca5'),
  "emp_id": 186,
  "name": 'Sneha',
  "department": 'HR',
  "designation": 'Manager',
  "city": 'Delhi',
  "salary": 78000,
  "experience": 7
},
{
  "_id": ObjectId('699adal77dfa6af86b628ca5'),
  "emp_id": 186,
  "name": 'Sneha',
  "department": 'HR',
  "designation": 'Manager',
  "city": 'Delhi',
  "salary": 78000,
  "experience": 7
}
}
```

```

{
  _id: ObjectId('699ada177dfa6af86b628ca0'),
  emp_id: 189,
  name: 'Nara',
  department: 'IT',
  designation: 'Developer',
  city: 'Malkata',
  salary: 65000,
  experience: 0
},
{
  _id: ObjectId('699ad84d635e9eb4f2628ca0'),
  emp_id: 181,
  name: 'Amit',
  department: 'IT',
  designation: 'Developer',
  city: 'Malkata',
  salary: 60000,
  experience: 3
},
{
  _id: ObjectId('699ada177dfa6af86b628ca0'),
  emp_id: 181,
  name: 'Amit',
  department: 'IT',
  designation: 'Developer',
  city: 'Malkata',
  salary: 60000,
  experience: 3
},
{
  _id: ObjectId('699ad84d635e9eb4f2628ca3'),
  emp_id: 184,
  name: 'Pooja',
  department: 'Finance',
  designation: 'Accountant',
  city: 'Chennai',
  salary: 55000,
  experience: 0
},
{
  _id: ObjectId('699ada177dfa6af86b628ca3'),
  emp_id: 184,
  name: 'Pooja',
  department: 'Finance',
  designation: 'Accountant',
  salary: 55000,
  experience: 0
}
]

```

Q8. Display top 3 highest paid employees.

Query: `db.student_CSE_326.find().sort({salary:-1}).limit(3)`

Output:

```

[
  {
    _id: ObjectId('699ada177dfa6af86b628ca8'),
    emp_id: 189,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  },
  {
    _id: ObjectId('699ad84d635e9eb4f2628ca8'),
    emp_id: 189,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  },
  {
    _id: ObjectId('699ada177dfa6af86b628ca7'),
    emp_id: 188,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  }
]

```

Query: db.student_CSE_326.find({experience:{\$gt:5}})

```
{
  _id: ObjectId('699ad84d635e9eb4f2628ca5'),
  emp_id: 185,
  name: 'Karan',
  department: 'IT',
  designation: 'Developer',
  city: 'Kolkata',
  salary: 65000,
  experience: 6
},
{
  _id: ObjectId('699ad84d635e9eb4f2628ca5'),
  emp_id: 186,
  name: 'Sneha',
  department: 'HR',
  designation: 'Manager',
  city: 'Delhi',
  salary: 70000,
  experience: 7
},
{
  _id: ObjectId('699ad84d635e9eb4f2628ca7'),
  emp_id: 188,
  name: 'Meera',
  department: 'IT',
  designation: 'Developer',
  city: 'Bangalore',
  salary: 75000,
  experience: 8
},
{
  _id: ObjectId('699ad84d635e9eb4f2628ca8'),
  emp_id: 189,
  name: 'Rohit',
  department: 'IT',
  designation: 'Team Lead',
  city: 'Pune',
  salary: 90000,
  experience: 10
},
{
  _id: ObjectId('699ad177dfaf6a8f88b628ca4'),
  emp_id: 185,
  name: 'Karan',
  department: 'IT',
  designation: 'Developer',
  city: 'Kolkata',
  salary: 65000,
  experience: 6
},
{
```

Q10. Analyze query performance using explain().

Query: db.student_CSE_326.find({department:"IT"}).explain()

Output:

```
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'college_CSE_326.student_CSE_326',
    parsedQuery: { department: { '$eq': 'IT' } },
    indexFilterSet: false,
    queryHash: 'EB9C5E66',
    planCacheShapeHash: 'EB9C5E66',
    planCacheKey: 'BF9F2F91',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: { department: { '$eq': 'IT' } },
      direction: 'forward'
    },
    rejectedPlans: []
  },
  queryShapeHash: 'FECTEED837FC4832D8C50F5AF8CEA6F8180E145483EDC78E07EA831BC4722EAD',
  command: {
    find: 'student_CSE_326',
    filter: { department: 'IT' },
    '$db': 'college_CSE_326'
  },
  serverInfo: {
    host: 'LAPTOP-DKSHWIG3',
    port: 27817,
    version: '0.2.3',
    gitVersion: '36f41c9c38a2f13f834d033ba03c3463c091f001'
  },
  serverParameters: {
    internalQueryFacetBufferSizeBytes: 104857600,
    internalQueryFacetMaxOutputDocSizeBytes: 104857600,
    internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
    internalDocumentSourceGroupMaxMemoryBytes: 104857600,
    internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
    internalQueryProhibitBlockingMergeOnMongoS: 0,
    internalQueryMaxAddToSetBytes: 104857600,
    internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
    internalQueryFrameworkControl: 'trySseRestricted',
    internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
  },
  ok: 1
}
```

```
_id: ObjectId('699ad84d635e9eb4f2628ca0')
emp_id : 101
name : "Amit"
department : "IT"
designation : "Developer"
city : "Kolkata"
salary : 60000
experience : 3
```

```
_id: ObjectId('699ad84d635e9eb4f2628ca1')
emp_id : 102
name : "Neha"
department : "HR"
designation : "HR Executive"
city : "Delhi"
salary : 45000
experience : 2
```

```
_id: ObjectId('699ad84d635e9eb4f2628ca2')
emp_id : 103
name : "Ravi"
department : "IT"
designation : "Tester"
city : "Mumbai"
salary : 50000
experience : 4
```

```
_id: ObjectId('699ad84d635e9eb4f2628ca3')
emp_id : 104
name : "Pooja"
department : "Finance"
designation : "Accountant"
city : "Chennai"
salary : 55000
experience : 5
```

```
_id: ObjectId('699ad84d635e9eb4f2628ca4')
emp_id : 105
name : "Karan"
department : "IT"
designation : "Developer"
city : "Kolkata"
salary : 65000
experience : 6
```

```
_id: ObjectId('699ad84d635e9eb4f2628ca5')
emp_id : 106
name : "Sneha"
department : "HR"
designation : "Manager"
city : "Delhi"
salary : 70000
experience : 7
```

```
_id: ObjectId('699ad84d635e9eb4f2628ca6')
emp_id : 107
name : "Arjun"
department : "Finance"
designation : "Analyst"
city : "Mumbai"
salary : 48000
experience : 2
```

```
_id: ObjectId('699ad84d635e9eb4f2628ca7')
emp_id : 108
name : "Meera"
department : "IT"
designation : "Developer"
city : "Bangalore"
salary : 75000
experience : 8
```

```
_id: ObjectId('699ad84d635e9eb4f2628ca8')
emp_id : 109
name : "Rohit"
department : "IT"
designation : "Team Lead"
city : "Pune"
salary : 90000
experience : 10
```

```
_id: ObjectId('699ad84d635e9eb4f2628ca9')
emp_id : 110
name : "Anjali"
department : "HR"
designation : "Recruiter"
city : "Kolkata"
salary : 42000
experience : 1
```


